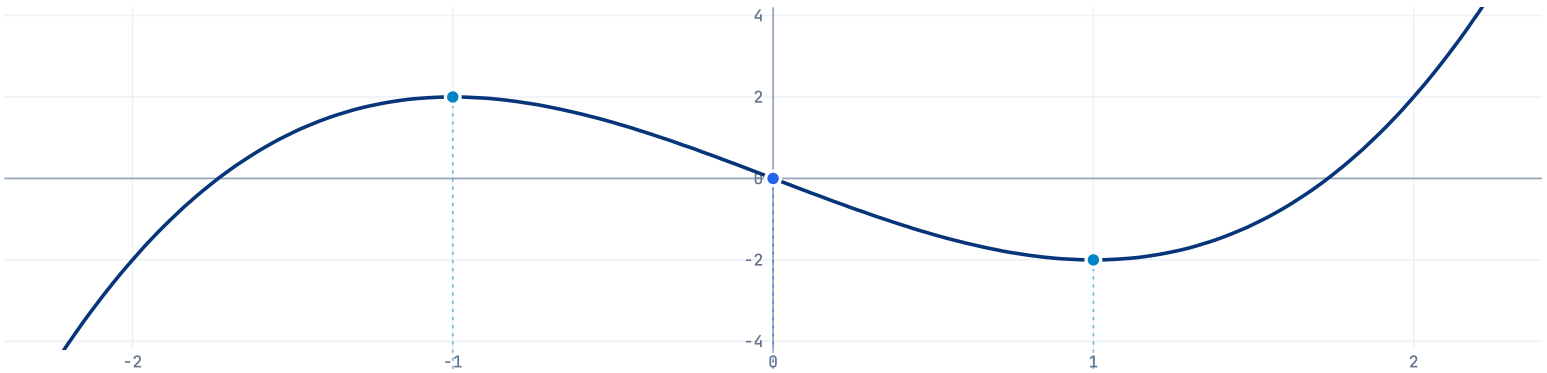
FIRST AND SECOND DERIVATIVE

Shape of a cubic from its derivatives

$f'(x) = 3x^2 - 3,\ f''(x) = 6x$

2 CRITICAL POINTS · 1 INFLECTION



SIGN OF F'	+	−	+
SIGN OF F''	− concave down		+ concave up

KEY POINTS

1 LOCAL MAXIMUM
$(-1, 2)$

2 INFLECTION
$(0, 0)$

3 LOCAL MINIMUM
$(1, -2)$

GLOBAL

CRITICAL POINTS
$f'(x) = 0$ at $x = \pm 1$

DIRECTION
rises, falls, rises
from the sign of f'

BENDING
down, then up
from the sign of $f'' = 6x$